

Mohamed Abdirizak Hassan

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PROFESSIONAL SUMMARY

Analytical and impact-driven professional with a First-Class Honours background in Computer Science, experienced at the intersection of data science, digital systems, and data-driven enterprise development. Skilled in leveraging descriptive analytics, database technologies (SQL, Python), and data visualization to train entrepreneurs on empirical business management, improve financial inclusion, and drive community-led economic resilience in complex, resource-constrained environments.

EDUCATION

Bachelor of Science in Computer Science | University of Nairobi **09/2021 – 09/2025**

- Grade: First Class Honours
- Honours/Scholarships: DAFI (Albert Einstein German Academic Refugee Initiative) Scholar
- Relevant Coursework: Distributed Databases, Business Intelligence & Analytics, Data Structures & Algorithms, Machine Learning, Probability & Statistics

PROFESSIONAL EXPERIENCE

Business Development Advisor | Inkomoko – Dadaab, Kenya **06/2026 – PRESENT**

- Quantitative Data Analysis & Financial Modeling: Performed exploratory and descriptive data analysis on client operational datasets, modeling key performance indicators (KPIs) such as gross margin distribution, inventory turnover rates, unit economics, and cash-flow to evaluate business viability.
- Data-Driven Training & Capacity Building: Designed and delivered data literacy and business management training modules (in Somali, Swahili, and English), teaching non-technical entrepreneurs how to capture daily transactional logs and interpret basic financial data to drive informed purchasing and pricing decisions.
- Data-Driven Consulting & Forecasting: Conducted regular on-site consulting sessions utilizing historical sales and expense data to identify operational leakages and formulate data-backed growth and risk-mitigation strategies.
- Data Verification & Credit Underwriting Support: Cleaned, validated, and synthesized disparate client data streams (intake surveys, site visit data, and site verification metrics) to generate auditable data profiles, directly supporting credit scoring, due diligence, and catalytic loan disbursements.

Data Engineer, KilimoVision | Dadaab, Kenya **06/2025 – 05/2026**

- Developed, customized, and maintained Odoo backend modules to support business processes and client requirements using XML and Python.
- Designed complex database models, implemented server-side logic, and optimized backend operations for scalability.
- Integrated Odoo with third-party APIs and external systems, ensuring seamless data exchange.

- Troubleshoot, debugged, and resolved issues within the Odoo framework to ensure high reliability.
- Modeled complex PostgreSQL schemas and query indexes for high concurrency.

Full Stack Engineer | Skill Up Technology – Nairobi, Kenya

10/2024 – 04/2026

- Architected and maintained secure REST API data ingestion pipelines using Python and Node.js, orchestrating continuous, idempotent data synchronization and file object storage with AWS S3 and Cloudflare R2.
- Designed end-to-end data preparation, chunking, and embedding pipelines to power AI chatbot integrations via Retrieval-Augmented Generation (RAG), implementing validation checks to ensure high retrieval accuracy and source attribution.
- Optimized backend data queries and API response times by 25% through intelligent caching layers, pre-rendering strategies, and efficient batching mechanisms.
- Containerized applications and automated data workflows using Docker, deploying resilient cloud infrastructure across AWS, Vercel, and Cloudflare with automated CI/CD build/test pipelines.
- Led and mentored a squad of 4 engineers, conducting rigorous code reviews and pairing sessions to enforce clean architecture, automated testing coverage, and "pipeline-as-code" documentation standards.

RESEARCH & DATA PROJECTS

Distributed Database System

***Tech Stack:** MySQL, PostgreSQL, Python, SQL, Shell Scripting, Docker*

- Designed and implemented a distributed database system across heterogeneous environments using vertical and horizontal fragmentation techniques.
- Automated fragmentation, distribution, and reconstruction of data using Python scripts and SQL queries.

AI-Powered Crop Diagnosis Pipeline (Final Year Project | UoN)

***Tech Stack:** Python, TensorFlow, Kotlin, Jetpack Compose, Firebase, Cloudflare R2*

- Developed an AI-driven mobile solution to diagnose crop diseases from smartphone-captured images using TensorFlow and MobileNetV2.
- Preprocessed datasets, trained custom CNNs, and deployed lightweight models suitable for on-device inference.

TECHNICAL SKILLS

- Languages & Scripting: Python (Advanced), SQL (PostgreSQL, MySQL), Bash, XML
- Data Engineering & Pipelines: ETL/ELT design, Idempotent loads, Incremental loading, Schema modeling
- Data Quality & Testing: PyTest, Automated quality gates, Freshness checks, Data contracts
- Transformation & Orchestration: dbt, Orchestration (Dagster/Airflow concepts), REST API ingestion
- Databases: PostgreSQL (Optimization & Indexing), MySQL, MariaDB, AWS S3, Cloudflare R2
- DevOps & Engineering: Git/GitHub, Docker, CI/CD, Linux, Pipeline-as-Code
- Data Governance: RBAC, Anonymization, Audit logging, Humanitarian data protection standards